Olist Brazilian E-Commerce Data Analysis

A comprehensive data engineering and analytics project by **DEPI, ITC Online Group 3, Team 3**. From raw Kaggle dataset to interactive Power BI dashboard analyzing **96.5K orders** across Brazil — January 2017 through August 2018.

96.5K

Orders Analyzed

93.4K

Unique Customers

15.4M

Total Revenue (BRL)

4

Core DB Tables

Database Design

Normalized 4-table SQL schema from 9-table source

Data Engineering

Python ETL pipeline with UTF-8 encoding and validation

Power BI Dashboard

15+ DAX measures across 3 analytical pages

Analytical Insights

Revenue, customer retention, and delivery performance

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Project Objective & Dataset

Primary Goal

Transform raw e-commerce data into a structured relational database and build an interactive Power BI dashboard to analyze sales performance, customer behavior, and delivery operations across Brazil.

Scope of Work

- Database design and normalization
- Data cleaning and ETL pipeline
- Power BI measures and dashboard development
- Insights on revenue, customer patterns, and delivery

Dataset: Olist Brazilian E-Commerce

Sourced from **Kaggle** — a public dataset covering **100K orders** with full order lifecycle data including customers, orders, items, products, payments, and geographic dimensions across multiple Brazilian states.

Why This Dataset?

Complete Lifecycle

Purchase → payment → delivery → review captured end-to-end

Geographic Depth

State and city-level data enabling regional performance analysis

Temporal Range

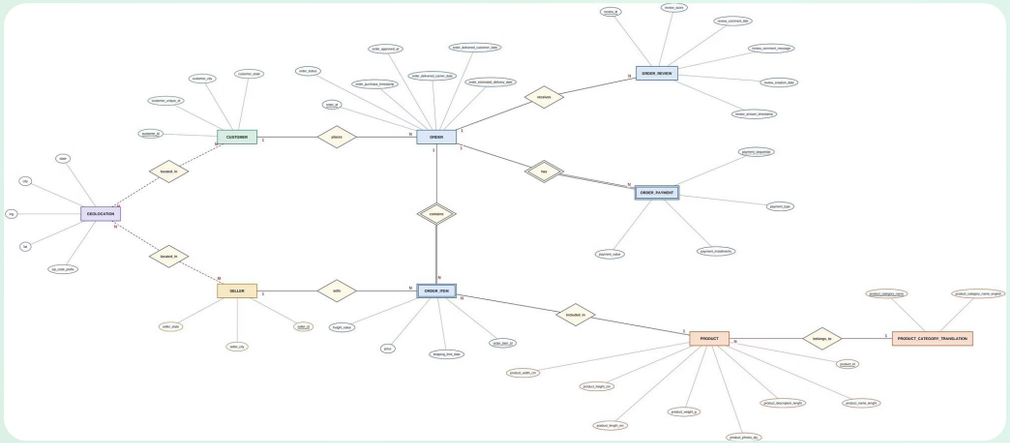
20-month window (Jan 2017–Aug 2018) supporting trend analysis

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Data Discovery & Database Design

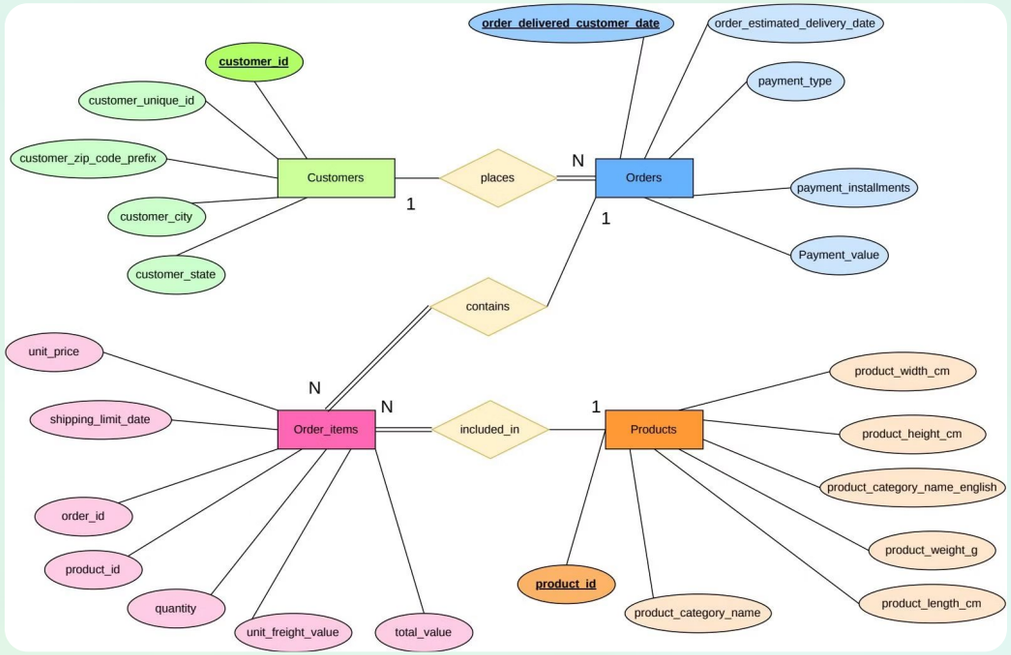
Original Dataset — Chen ERD (9 Tables)

The source schema comprised 9 interconnected tables: Customers, Orders, Order Items, Products, Payments, Reviews, Sellers, and Geolocation — representing the full operational data model of the Olist marketplace.



Target Analytics Schema (4 Tables)

Optimized for analytical queries by selecting the 4 core tables: **Customers, Orders, Order_items, Products**. Key design decision: payment_installments converted from numeric to Boolean to reduce redundancy and simplify aggregation logic.



Trade-off: Simplified schema improves query performance but excludes seller, review, and geolocation granularity from the analytical layer.

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Data Cleaning & SQL Implementation

1

Cleaning Operations

- Removed empty rows and duplicate records
- Converted all text fields to **UTF-8 encoding**
- Standardized column names and corrected data types
- Flagged suspicious records for manual review

2

Relational Validation

- Verified referential integrity: orders → customers, items → products
- Checked date consistency across order lifecycle fields
- Validated payment totals against line-item sums
- Confirmed geographic accuracy of state/city codes

3

SQL Schema Design

- Four normalized tables with **PRIMARY KEY** and **FOREIGN KEY** constraints
- **NOT NULL** constraints on critical fields
- Python notebooks for ETL with parameterized queries
- Automated pipeline from raw CSV to loaded database

4

Core Lesson

Data quality requires validating **relationships between tables** and consistency across temporal and geographic dimensions — not just cleaning individual cells in isolation.

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Power BI Measures & Calendar

Calendar Dimension Table

A continuous date dimension spanning **Jan 2017 – Aug 2018** was constructed to enable time-based slicing and aggregations. Attributes include Year, Month, Quarter, and Day of Week — essential for time-intelligence functions in DAX.

Core KPI Measures

Revenue

Total, Product Sales, Freight Value

Volume

Delivered Orders, Units Sold

Customers

Unique Customers, Avg Order Value, Revenue per Customer

Advanced & Time-Intelligence Measures

YTD Sales

Year-to-date cumulative revenue using DATESYTD

Month-over-Month %

MoM growth with guard clause for low-volume periods to prevent misleading spikes

Delivery Performance

Average Delivery Days, Late Delivery Rate, On-Time Rate

Customer logic: Unique customers use `customer_unique_id`; repeat customers defined as those with more than 1 order on record.

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Power BI Dashboard: Executive Overview

15.4M

Total Revenue (BRL)

13.2M product + 2.2M freight

96.5K

Delivered Orders

93.4K

Unique Customers

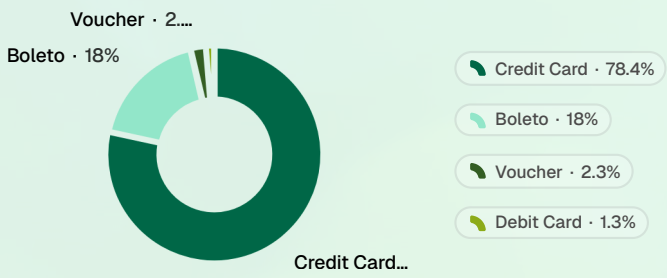
137

Avg Order Value (BRL)

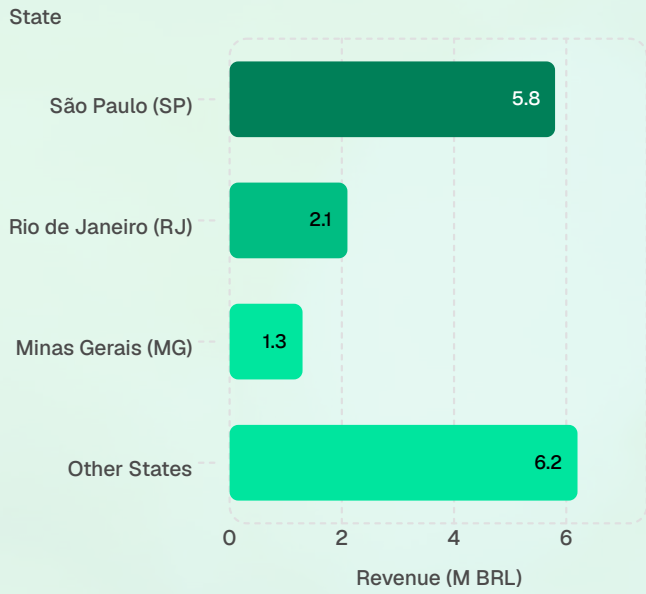
Monthly Revenue Trend

Growth from **0.1M (Jan 2017)** to a peak of **1.2M (Nov 2017)**, then stabilizing at approximately **1.0M/month** through Aug 2018 — indicating market maturation after initial rapid growth.

Payment Methods



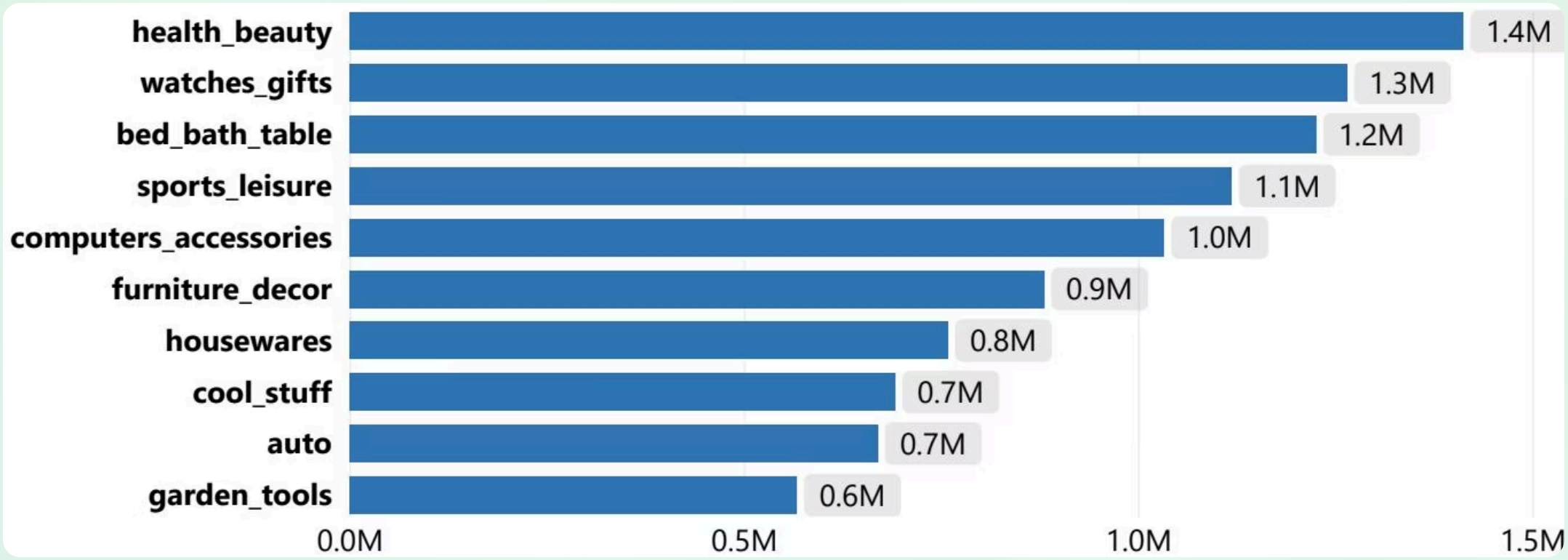
Geographic Concentration



Top 3 states (SP, RJ, MG) account for approximately **60%** of total revenue.

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Products & Customers Page



Product KPIs

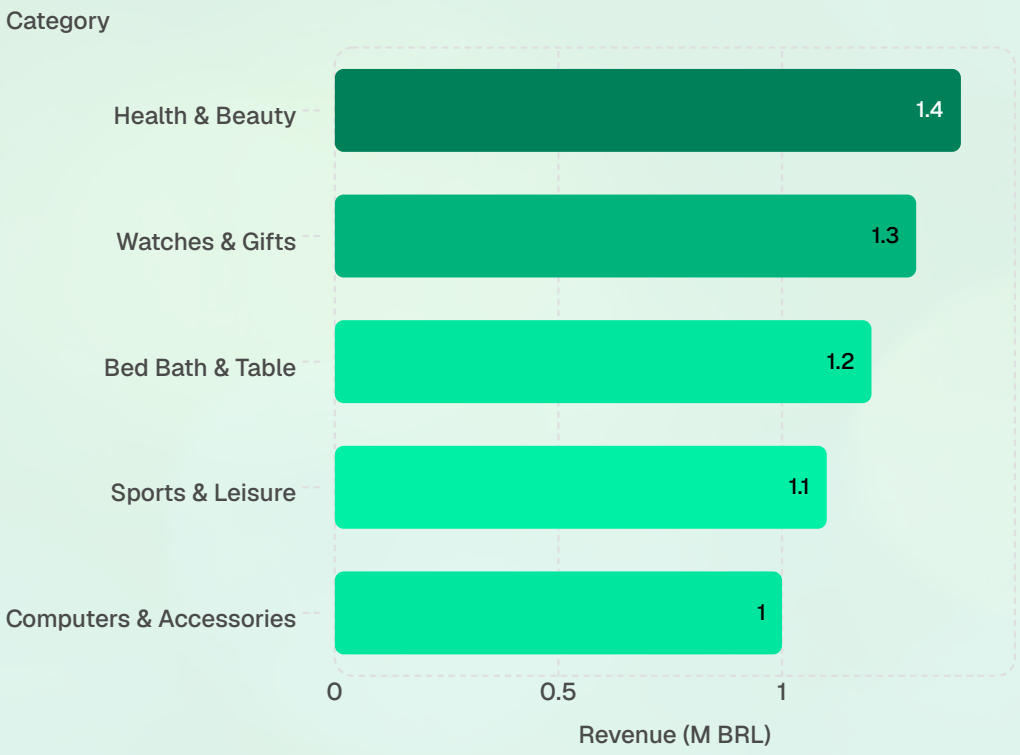
74

Product categories

110K

Units sold

Top 5 Revenue Categories



Top 5 categories account for approximately **39%** of total revenue.

Customer KPIs

165

Revenue / Customer (BRL)

3%

Repeat Customer Rate

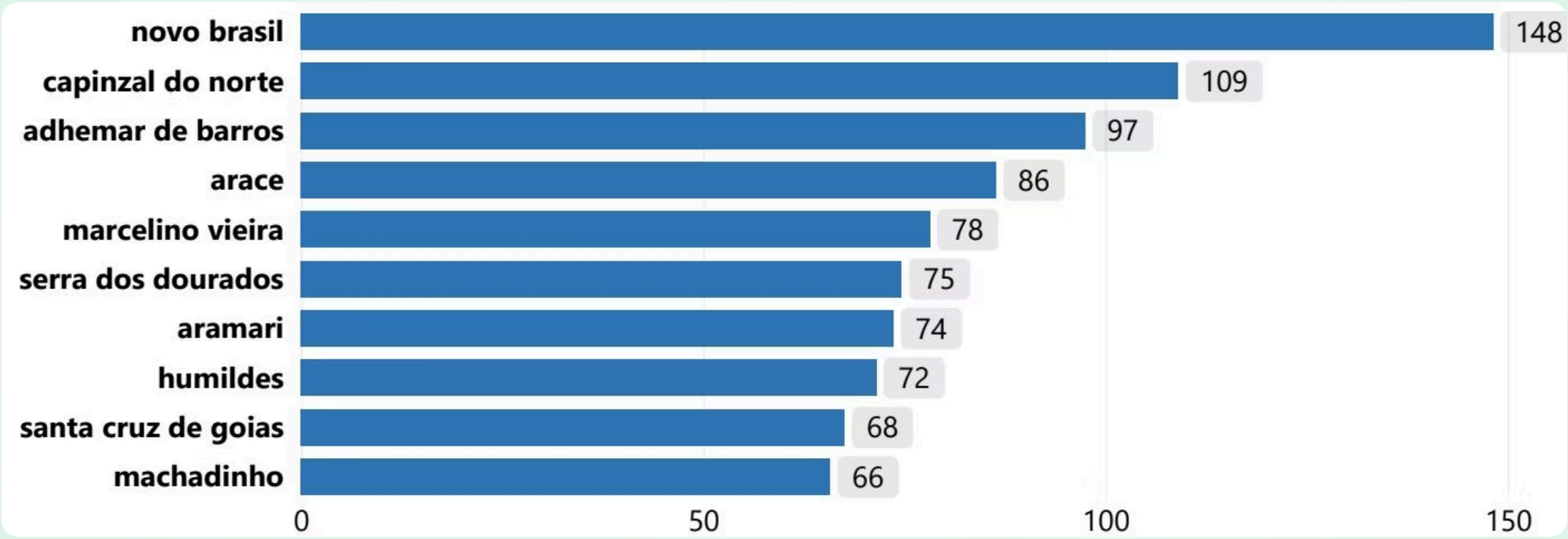
Geographic Customer Concentration

São Paulo city alone accounts for **15.5K customers**. The top 10 cities represent approximately **45%** of the entire customer base — mirroring the revenue concentration pattern observed at the state level.

Strategic insight: A 3% repeat rate signals a predominantly transactional customer base. Retention programs targeting high-value segments could materially improve lifetime value metrics.

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Operations & Delivery Page



12.6

Avg Delivery Days

91.9%

On-Time Rate

8.1%

Late Delivery Rate

7.8K

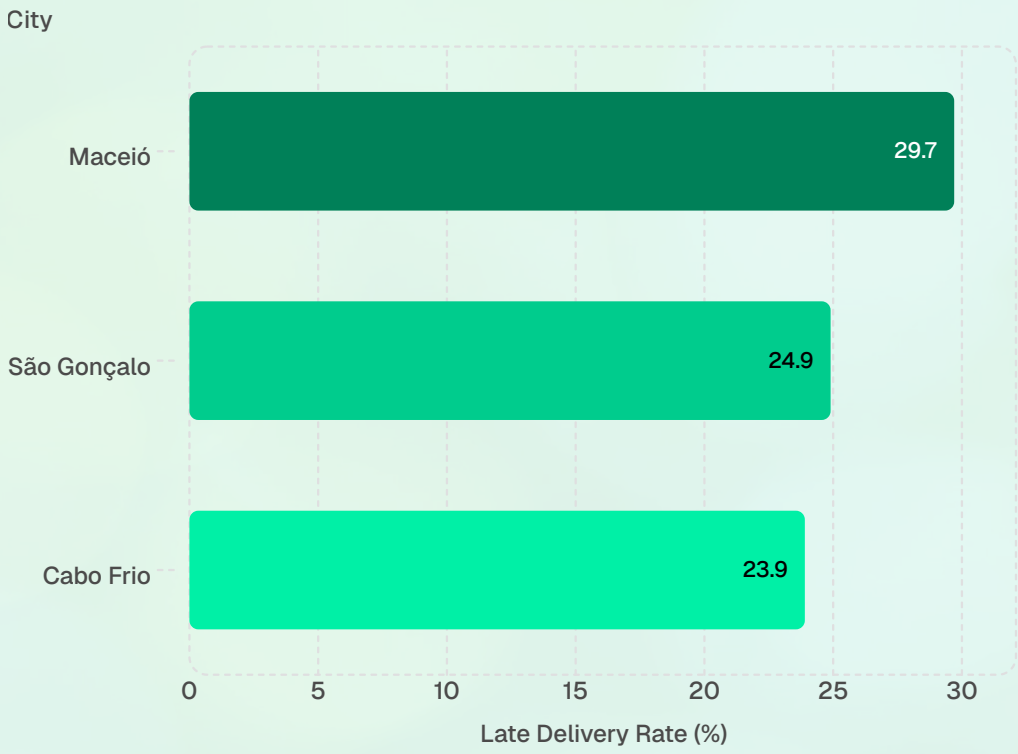
Late Orders

Geographic Delivery Disparity

Remote municipalities exhibit extreme delivery delays relative to urban centers. Observed outliers include **Novo Brasil (148 days)** and **Capinzal do Norte (109 days)** — likely reflecting last-mile logistics constraints in underserved regions rather than systemic carrier failure.

⚠ Small municipalities with few orders produce statistically unreliable averages. A minimum sample threshold of **≥10 orders** is recommended before drawing operational conclusions.

Late-Delivery Hotspots



These urban hotspots suggest **carrier or logistics challenges** rather than geographic remoteness. Late delivery may also reflect inaccurate delivery estimates or address data issues — causality requires further investigation.

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Key Findings & Limitations

Descriptive Findings

Revenue Concentration

São Paulo = **38% of revenue**. Geographic diversification could reduce concentration risk and unlock growth in underserved states.

Customer Dynamics

93.4K unique customers, **3% repeat rate**, avg order value **137 BRL**. Low loyalty signals a retention opportunity — not a confirmed churn problem.

Delivery Performance

91.9% on-time overall, but extreme geographic variation. Late delivery ≠ carrier fault; may reflect inaccurate estimates or address data quality issues.

Analytical Limitations

Scope Reduction

4 of 9 source tables selected; seller, review, and geolocation data excluded from the analytical layer.

Boolean Conversion

`payment_installments` converted from numeric to Boolean — installment-count granularity is lost.

Small-Sample Bias

Niche categories and small municipalities produce unreliable averages; minimum thresholds required.

Identity Accuracy

Repeat-customer definition depends on `customer_unique_id` accuracy — unverifiable from public data.

All findings are descriptive observations and hypotheses. No causal relationships are claimed without further controlled analysis.

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Team Contributions & Deliverables

Mohamed Sokarno *(Lead)*

- Dataset discovery and selection
- SQL schema design and data loading
- UTF-8 encoding resolution
- Project review and documentation

Mahmoud Al-Nabawi

- Source ERD in Chen notation
- Data dictionary compilation

Mohamed El-Gml

- Python data cleaning and preparation
- ETL notebook development

Mahmoud Saqr

- Calendar table construction
- 15+ Power BI KPI measures
- Time-intelligence DAX functions

Mostafa Salem

- Four-page Power BI dashboard
- Executive Overview, Products & Customers, Operations & Delivery pages

Project Deliverables

SQL Database

96.5K orders, 93.4K customers

Python ETL Pipeline

Automated cleaning and loading

Power BI Dashboard

3 analytical pages, 15+ measures

Documentation

ERD, data dictionary, insights report

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Thank you

& Questions

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